INVESTIGATION OF THE DYNAMICS OF TRIPLE SYSTEMS BY THE MONTE CARLO METHOD. III. THE CASE OF COMPONENTS WITH DIFFERENT MASSES

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The equations of motion for triple systems were integrated numerically on a computer for 300 different initial configurations. The components were initially at rest, and their masses were as follows: 1) $m_1 = 9$, $m_2 = 3$, $m_3 = 1$; 2) $m_1 = 3$, $m_2 = 3$, $m_3 = 1$; 3) $m_1 = 3$, $m_2 = 1$, $m_3 = 1$. In all cases, the motion of the components terminated in decay of the system. In each case, the decay occurred after a close triple approach of the components. The mean decay time is $\overline{T} = (27.8 \pm 3.5)\tau$, where τ is the mean time of motion of a component through the system. In the great majority of cases the triple system loses the component with the smallest mass. The spread in the masses of the components reduces the decay time by a factor of about three.

A study of the dynamics of triple systems with components of equal mass was reported in [1] and [2]. The investigation was carried out by computer integration of the equations of motion for the components of triple systems with randomly specified initial configurations. In 100 cases the initial velocities were assumed in [1] to have had small values in the plane of the configurational triangle, and in 100 further cases in [2] the components were assumed initially at rest.

In all the cases investigated, the triple system decayed after close approach of the components. The mean decay time was $\overline{T} = (95.4 \pm 6.9)\tau$, where τ is the mean time spent by each component in the system (see [1] for the definition of τ).

As in [1] and [2], in the present investigation we have carried out a numerical integration of the equations of motion of the triple-system components, but the masses of the components were assumed to be different. The following mass combinations were investigated: 1) $m_1 = 9$, $m_2 = 3$, $m_3 = 1$; 2) $m_1 = 3$, $m_2 = 3$, $m_3 = 1$; 3) $m_1 = 3$, $m_2 = 1$, $m_3 = 1$. The system of units is the same as in [1]. The components were assumed to be initially at rest. In the case of equal masses, all the possible configurations of the triple systems lie in the domain D in Fig. 1 which is bounded by the coordinate axes and a circular arc of unit radius, drawn from the point ($\xi = -1/2$, $\eta = 0$). In the case of different masses, the domain D does not include all the possible configurations. The new domain is the entire half-plane bounded by the ξ axis. On the other hand, the distances between the components of the system must not exceed 10 units of the distance at which it is assumed that the system decays by definition [1]. Let us therefore consider a new domain D' of all the possible configurations of triple systems, defined by a semicircle of radius 10 units of distance and centered on the origin.

To construct the initial configurations of the triple systems with components of different mass, we have taken the pseudorandom numbers ξ and η from [1], and used the formulas transforming D into a region not very different from D' to calculate the coordinates ξ', η', specifying the random initial position of the point C. The values of the coordinates ξ', η' are given in Table 1. When the mass ratios are 9, 3, 1 and 3, 1, 1, the coordinate axis is not an axis of symmetry of the domain D' and, therefore, when the initial conditions of the triple systems are selected from Table 1, only the first 50 pairs of numbers ξ' and η' are taken, and to each value of ξ' we assign first a positive sign and then a negative sign. For triple systems with masses $m_1 = 3$, $m_2 = 3$, and $m_3 = 1$, the coordinate axis is the axis of symmetry of the region of all the possible initial configurations of the triple system and, therefore, all the 100 coordinate pairs are taken for this mass ratio.

For each mass ratio (9, 3, 1; 3, 3, 1; and 3, 1, 1) we calculated the component trajectories for 100 different initial configurations. The equations of motion were linearized before the trajectories were calculated. The calculations were checked as in [1] and [2]. The motion of the components in all the systems was terminated by an exact or conditional decay of the triple systems.

In contrast to the decay criterion in [1, 2], where the masses of the components were such that $m \geq m_i$, $m \geq m_j$ (m is the mass of the departing component, m_i and m_j are the masses of the close components), which is valid for equal masses of the components, in the present study we have used the criterion given by Tevzadze [3].

According to this criterion, if at the initial time t_0

$$E<0,\ \rho(t_0) \geqslant \frac{2Gm^2}{3|E|},\ \text{and}$$

$$\rho(t_0)^2 > \frac{2GM}{m_i + m_j}\left[\frac{m_i}{\rho(t_0) - g} + \frac{m_j}{\rho(t_0) - q}\right], \quad (1)$$

where $g = m_i r_{max}/(m_i + m_j)$, $q = m_j r_{max}/(m_i + m_j)$, then $\rho(t) \to \infty$ as $t \to \infty$. In these expressions E is the total energy of the system, G is the gravitational constant, M is the mass of the components, r_{max} is the maximum possible distance between close components, and $\rho(t)$ is the distance of the distant component from the center of mass of the close components. The values of r_{max} have been given by Merman [4]:

$$r_{max} = G\frac{\sum_{i,j=1}^{3} m_i m_j}{|E|}. \quad (2)$$

Table 1

no.	ξ'	η'	no.	ξ'	η'
1	2	3	4	5	6
1	1.530844530	0.281758510	51	9.897502790	0.351546184
2	6.306644850	0.271786951	52	3.969836761	0.516086826
3	0.586409706	0.376641268	53	9.018280581	2.505227760
4	0.438322352	0.499630538	54	3.067183881	0.184321248
5	8.849380451	0.021331006	55	0.036530617	3.052396969
6	1.169632410	0.121508198	56	0.157343465	0.837634053
7	3.878655622	0.651987233	57	2.407748181	1.000608162
8	1.466861089	0.626014310	58	0.039323796	5.397602950
9	5.376967151	0.722344918	59	1.254814240	2.178333211
10	0.700272955	2.942157730	60	0.423804407	0.696777909
11	0.061019896	2.309376941	61	3.538867271	2.428183280
12	0.058518012	1.461777710	62	3.400479221	6.058513182
13	0.524178883	0.094018993	63	1.436241990	0.154822977
14	1.132229506	3.474061231	64	0.234115877	1.565074001
15	8.268518520	0.132728839	65	0.421050581	1.254845370
16	3.036166130	0.793502270	66	1.525865330	1.112525601
17	2.110731741	0.083115956	67	1.743369381	2.178891880
18	7.700283291	0.287837173	68	0.642880812	0.179802891
19	4.749117880	0.778826453	69	0.178184193	1.444668011
20	0.247164822	1.105720741	70	0.156767387	8.655581531
21	1.607084241	0.018175596	71	0.176755433	0.815456257
22	1.967298481	1.917649752	72	1.546244840	0.064848318
23	2.601429880	6.345181961	73	5.171173061	2.867921350
24	0.144089332	6.430309540	74	6.988977290	0.127183530
25	0.820856972	1.372113641	75	1.306991751	2.230054082
26	2.494591261	0.095394165	76	0.535627260	5.885260821
27	1.367592220	2.617918441	77	2.190805521	6.421980102
28	2.166068390	0.184773944	78	1.854081401	0.037135860
29	1.398423571	0.405256122	79	1.244812341	0.274200596
30	0.002553481	1.411038982	80	2.207584111	0.645684734
31	2.220389780	1.130497220	81	0.734044645	1.815467440
32	3.284820620	0.178978249	82	1.681666701	0.653482208
33	4.320537641	1.089135821	83	2.047956540	0.428628578
34	0.850873779	0.171355445	84	1.170159791	7.824601480
35	0.049774664	2.893940731	85	7.526092770	0.506527494
36	2.412755611	3.704948852	86	2.810004181	4.331742602
37	1.009304771	0.031562542	87	0.008901173	1.507866480
38	0.773147115	0.209885466	88	0.045691683	0.976742777
39	0.031505655	2.675449670	89	4.329382450	0.975187567
40	0.017229832	1.730411361	90	0.769523497	1.060377371
41	0.436759069	1.832797490	91	1.043539051	1.622453740
42	0.001521472	1.919225650	92	1.043242290	0.629017490
43	1.041762921	0.040544395	93	0.070623741	0.388412661
44	0.951275538	0.154375299	94	0.016289413	0.013719886
45	0.413421343	1.031453551	95	1.702270990	0.454068623
46	0.874035866	2.841184390	96	0.027074829	2.019959990
47	0.328774180	0.330854350	97	0.273074279	1.362998941
48	4.410341941	0.196363674	98	1.169154601	0.259721043
49	3.845804090	2.183368491	99	0.660164106	7.013163491
50	2.430467770	0.518036195	100	8.391696810	0.347129432

Table 2

no.	m_i	1, 1, 1		9, 3, 1	3, 3, 1	3, 1, 1
1	$\overline{T}$	95.4±6.9		26.8±4.2	27.4±3.0	29.2±3.4
2	n	34.		34	41	24
3	n^*	163		98	94	97
4	N_1	150		97	90	99
5	N_2	118		94	65	92
6	N_3	24		29	10	35
7	N_A	—		0	3	0
8	N_B	—		10	3	52
9	N_C	—		90	94	48
10	$0<P\leqslant 0.2$	n_1	125	4	5	6
11		$\left(\frac{\overline{\Delta' E}}{\overline{-E}}\right)$	1.273	1.652	0.127	19.627
12	$0.2<P\leqslant 0.5$	n_2	75	21	23	31
13		$\left(\frac{\overline{\Delta' E}}{\overline{-E}}\right)$	0.279	5.713	2.460	3.036
14	$0.5<P\leqslant 1.0$	n_3	—	32	27	40
15		$\left(\frac{\overline{\Delta' E}}{\overline{-E}}\right)$	—	1.331	0.587	1.407
16	$P>1.0$	n_4	--	43	45	23
17		$\left(\frac{\overline{\Delta' E}}{\overline{-E}}\right)$	--	0.531	0.116	0.960

Table 2 gives our results for mass ratios 1, 1, 1 (see [1–2]); 9, 3, 1; 3, 3, 1; and 3, 1, 1. It includes the values of the mean decay time $\bar{T}$, the number n of triple systems in which the motion of the component

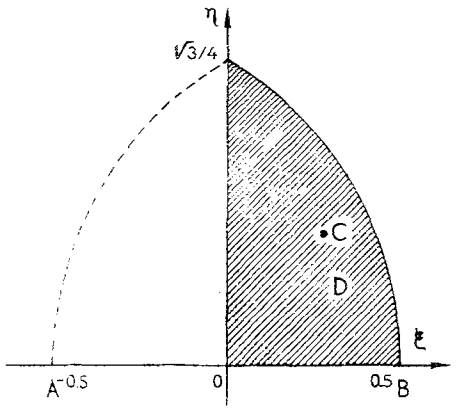


Fig. 1

terminates in the conventional decay, the number n* of decays occurring after the closest triple approach of the components, the numbers N_1, N_2, N_3 of systems for which the ratio of the energy change ΔE due to accumulation of errors during the computation to the total energy E of the system does not exceed 0.01, 0.005, and 0.0005, respectively, and the numbers N_A, N_B, N_C of systems with components of different masses in which components A, B, or C leave the system on decay, respectively. The next lines show the number of cases in which the perimeter P of the triangle at the time of triple approach resulting in decay lies in the indicated range, and the mean value $(\overline{\Delta' E/-E})$, where $\Delta' E$ is the energy excess of the escaping component over the energy demanded by the criterion (1).

It is clear from Table 2 that for the mass ratio of 1, 1, 1 the ratio $\Delta E/-E$ is relatively high since, for the first 100 cases, we have not used linearization and, moreover, the decay of triple systems with components of equal mass occurred later.

Our results lead us to the following conclusions: 1) for all values of the masses, the motion of the components ends in the decay of the triple system; the reduction in the value of $(\overline{\Delta' E/-E})$ with increasing P shows that the decay of the triple system is a real phenomenon and is not the result of an accumulation of computational errors; 2) the spread in the masses of the components increases the decay time of triple systems by a factor of about three (this conclusion appears to be valid for systems consisting of a large number of bodies); 3) in the great majority of cases the triple system is left by the component with the smallest mass; 4) the results of the present study confirm the conclusion reported in [1] and [2] that the closer the triple approach, the more probable is the decay of the triple system.

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